

VLADYSLAV HORBATENKO

🌐 rifolio.me | vladyslav.horbatenko.work@gmail.com | Copenhagen 2400, Denmark
🌐 https://www.linkedin.com/in/vladyslav-horbatenko/ | 🌐 https://github.com/rifolio

SKILLS

Technical Skills: Python, Java, MySQL, PostgreSQL, Git, Docker, REST API, Agentic LLMs
Frameworks: Django, Next.js, DSPy, Pydantic, LangChain, Celery, MCP
Other: AWS, GCP, Dokploy, Pandas, Numpy, Matplotlib, Seaborn, SciPy, Tensorflow, PyTorch, Scikit Learn, Pinecone, ChromaDB, LangGraph, LangSmith, HuggingFace, LaTeX

WORK EXPERIENCE

Co-Founder & CTO

PeerMind

Sep 2025 – Present

Denmark, Copenhagen

- Architected and shipped a full-stack AI-powered educational platform (Django, Next.js, PostgreSQL, Redis, Celery) end-to-end, with Test/Production CI/CD pipelines via GitHub Actions and Dokploy, Traefik reverse proxy, and Caddy for secure signed media serving.
- Built a multi-document preprocessing pipeline using VLMs, custom OCR and embedding models for semantic chunking, native document structure analysis, and multi-phase ToC detection, with LLM-based cross-document relationship analysis and merging.
- Implemented hybrid RAG with Pinecone vector search, agentic multi-query retrieval via DSPy, and GraphRAG-style page-span context assembly for high-fidelity content generation.
- Developed an adaptive learning system that analyzes per-student quiz interactions and mistakes, dynamically generating personalized remediation content (pages and quizzes) tailored to individual knowledge gaps.
- Built AI agents for both students and teachers. Multimodal AI Tutor with RAG search, image analysis, and streaming responses, and a ReAct page-editing agent with structured tool use, powered by LangChain and LangGraph with full LangSmith observability.
- Designed custom alerting systems to detect errors and anomalies in production, and a teacher analytics dashboard with engagement heatmaps, drop-off analysis, and content effectiveness tracking.

AI Engineer

Tryp.com

Sep 2024 – Present

Denmark, Copenhagen

- Developing and optimizing a LLM-based customer support: improved multi-turn dialogue handling, achieved 86% intent recognition accuracy, reduced average response time by 83%, and handled 74% of all inbound inquiries ($\approx 10,500$ /month).
- Built an agentic travel search agent capable of multi-step reasoning across destinations, pricing, and user constraints.
- Variety of internal automation for customer support team; helping to write emails in any language, receive and update guidelines, get notified.
- Developing a FNN model to estimate ticket-price change durations using over 70 million flight records.

Co-Founder (Lead AI Engineer)

CoreAI

Jan 2025 – Present

Denmark, Copenhagen

- Deployed an AI chatbot system based on AWS Bedrock, providing 24/7 multilingual support for Danish social help, higher education, and Swedish migration support, with other Nordic regions in development.
- Built a microservice-like AWS backend (API Gateway, Lambda, DynamoDB, LightSail) with a RAG pipeline for real-time responses and booking integration (EasyWeek API).
- Integrated Telegram and WhatsApp frontends via API Gateway & Lambda for fast user access.
- Co-authored the AWS Machine Learning Blog post “Build a scalable AI assistant to help refugees using AWS.”

Teacher Assistant

Roskilde University

Sep 2024 - Dec 2024

Denmark, Roskilde

- Guiding 52 students on debugging, problem-solving, and coding practices in a Java programming course.
- Providing feedback and grading on weekly homework assignments.

EDUCATION

Technical University of Denmark DTU: MSc in Human-Centered Artificial Intelligence

Sep 2025 - Jun 2027

Roskilde University RUC: BSc in Computer Science and Mathematics

Sep 2022 - Jun 2025